

ICT600 CORRECTED STUDENT SCHEME

July 2024

Question-ordered model answers and marking guide

Scope	July 2024 final examination
Arrangement	Answers follow the original paper from Question 1 through Part B. Coding, diagrams, and written responses are integrated once at their correct positions.
Assessment	100 marks: Part A 70, Part B 30
Status	Corrected student reference; equivalent technically correct answers may receive credit.

How to use this scheme

Question-order rule

Attempt the paper first. Then check the matching answer in this document. Each answer appears once, in the same order as the question paper.

- Equivalent correct wording, diagrams, algorithms, and code may receive full credit when they meet the stated requirements.
- For coding, award method marks when a minor syntax error does not invalidate an otherwise correct solution.
- For design or illustration questions, the labelled diagram is the model answer; prose alone is not treated as a complete design.

Part A

Question 1(a)

8 marks

Describe FOUR applications of HTML.

Mark allocation

Awardable element	Marks
Document structure	2
Links and embedded resources	2
Forms	2
Multimedia	2
Total	8

Complete model answer

- Structure web documents with headings, paragraphs, lists, and sections.
- Create hyperlinks and embed resources.
- Build forms for user input.
- Include multimedia such as images, audio, and video.

Question 1(b)

10 marks

Identify FIVE web-based CMS tools.

Mark allocation

Awardable element	Marks
First valid CMS	2
Second valid CMS	2
Third valid CMS	2
Fourth valid CMS	2
Fifth valid CMS	2
Total	10

Complete model answer

- WordPress, Blogger, Joomla, Drupal, and Wix or Weebly. Accept another correctly identified CMS/site-building platform.

Question 2(a)**6 marks****Describe SIX website-anatomy components.****Mark allocation**

Awardable element	Marks
Containing block	1
Logo	1
Navigation	1
Content	1
Footer	1
Whitespace	1
Total	6

Complete model answer

- Containing block, logo, navigation, content, footer, and whitespace, each with its function.

Question 2(b)**6 marks****Name SIX client-side development tools or tool categories.****Mark allocation**

Awardable element	Marks
First valid category or example	1
Second	1
Third	1
Fourth	1
Fifth	1
Sixth	1
Total	6

Complete model answer

- JavaScript libraries, front-end frameworks, web-application frameworks, programming languages, IDEs, and testing tools. Named products are acceptable when their category is clear.

Question 3(a)**8 marks****Classify FOUR mechanisms that enforce database restrictions.****Mark allocation**

Awardable element	Marks
Primary key and effect	2
Foreign key and effect	2
UNIQUE and effect	2
NOT NULL or CHECK and effect	2
Total	8

Complete model answer

- Primary key
- Foreign key
- UNIQUE constraint
- NOT NULL or CHECK constraint

Marker note

A plain non-unique index improves lookup performance but does not enforce a data restriction.

Question 3(b) - INSERT statement**10 marks****Insert all three ITEM rows with one complete SQL statement.****Mark allocation**

Awardable element	Marks
INSERT INTO table and column list	2
VALUES syntax	2
sh234/Adidas row	2
ts124/Nike row	2
cp456/Puma row	2
Total	10

Complete model answer

```
INSERT INTO ITEM (itemID, itemBrand, itemType, itemPrice, quantity)
VALUES
    ('sh234', 'Adidas', 'Shoe', 345.00, 2),
    ('ts124', 'Nike', 'T-Shirt', 99.99, 6),
    ('cp456', 'Puma', 'Cap', 35.75, 10);
```

Question 4(a) - PHP car array

4 marks

Use a PHP multidimensional array to display the three car rows.

Mark allocation

Awardable element	Marks
Correct nested array	2
Complete table output	2
Total	4

Complete model answer

```
<?php
$scars = [
    ['Proton', 15, 8],
    ['Perodua', 20, 10],
    ['Toyota', 10, 5],
];
?>
<table border="1">
  <tr><th>Name of Car</th><th>Stock</th><th>Sold</th></tr>
  <?php foreach ($scars as $scar): ?>
    <tr><td><?= $scar[0] ?></td><td><?= $scar[1] ?></td><td><?= $scar[2] ?></td></tr>
  <?php endforeach; ?>
</table>
```

Question 4(b) - exact PHP trace

6 marks

Give the output after each operation in the printed script.

Mark allocation

Awardable element	Marks
Length/original output	1
Uppercase	1
Lowercase	1
Replacement	1
Substring	1
Both strpos types/values	1
Total	6

Complete model answer

```
34
Covid19 is a Disease Outbreak News
COVID19 IS A DISEASE OUTBREAK NEWS
covid19 is a disease outbreak news
Covid19 is a Disease Outbreak Story
vid19 is a Disease Outbreak Story
int(21)
bool(false)
```

Question 5(a) - administrator access

6 marks

Allow only the administrator role to open user_view.php.

Mark allocation

Awardable element	Marks
Session started	1
Administrator condition	2
Error redirect for other users	2
exit and valid syntax	1
Total	6

Complete model answer

```
<?php
session_start();
if (!isset($_SESSION['role']) || $_SESSION['role'] !== 'admin') {
    header('Location: error.php');
    exit;
}
// Protected user_view.php content follows.
?>
```

Question 5(b) - logout

6 marks

Destroy the session and redirect when the supplied logout link is used.

Mark allocation

Awardable element	Marks
Session started	1
Logout parameter checked	1
Session cleared and destroyed	2
Redirect	1
exit/valid flow	1
Total	6

Complete model answer

```
<?php
session_start();
if (isset($_GET['logout'])) {
    $_SESSION = [];
    if (ini_get('session.use_cookies')) {
        $p = session_get_cookie_params();
        setcookie(session_name(), '', time() - 42000, $p['path'], $p['domain'],
            $p['secure'], $p['httponly']);
    }
    session_destroy();
    header('Location: login.php');
    exit;
}
?>
```

Part B

Part B Question 1 - patient-name analyser

15 marks

Provide both inputData.html and receiveData.php as a complete solution.

Mark allocation

Awardable element	Marks
Input page, textarea, action and POST	3
Receive, split and clean comma-separated names	3
Associative name-to-length array	3
Find longest name and length	4
Safe complete output	2
Total	15

Complete model answer

```
<!-- inputData.html -->
<form method="post" action="receiveData.php">
  <label>DATA: <textarea name="data" required></textarea></label>
  <button type="submit">SEND</button>
</form>

<!-- receiveData.php -->
<?php
$raw = trim($_POST['data'] ?? '');
$names = array_values(array_filter(array_map('trim', explode(',', $raw))));
$dataArray = [];
foreach ($names as $name) {
    $dataArray[$name] = mb_strlen($name);
}
arsort($dataArray);
$longestName = array_key_first($dataArray);

$longestLength = $longestName === null ? 0 : $dataArray[$longestName];
function e(string $v): string { return htmlspecialchars($v, ENT_QUOTES, 'UTF-8'); }
?>
<p>Longest name: <?= e($longestName ?? 'No name supplied') ?></p>
<p>Length: <?= $longestLength ?></p>
```

Part B Question 2 - PHP user registration

15 marks

Provide the full registration form and processing page.

Mark allocation

Awardable element	Marks
Form and controls	4
POST handling	2
Required-field validation	3
Success/error states	2
Safe encoded output	2
Valid structure	2
Total	15

Complete model answer

```
<?php
$values = ['fullname' => '', 'email' => '', 'birthdate' => '', 'gender' => ''];
$message = '';

function e(string $value): string {
    return htmlspecialchars($value, ENT_QUOTES, 'UTF-8');
}

if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    foreach ($values as $key => $unused) {
        $values[$key] = trim($_POST[$key] ?? '');
    }
    $password = $_POST['password'] ?? '';
    if ($values['fullname'] === '' || $values['email'] === '' || $password === '') {
        $message = 'Unsuccessful: enter all required information.';
    } else {
        $message = 'Successful signup for ' . $values['fullname'] . '.';
    }
}
?>
<!doctype html>
<html lang="en"><head><meta charset="utf-8"><title>User Signup</title></head>
<body>
<h1>User Signup</h1>
<form method="post">
    <label>Full name* <input name="fullname" value="<?= e($values['fullname']) ?>" required></label><br>
    <label>Email* <input type="email" name="email" value="<?= e($values['email']) ?>" required></label><br>
    <label>Password* <input type="password" name="password" required></label><br>
    <label>Birthdate <input type="date" name="birthdate" value="<?= e($values['birthdate']) ?>"></label><br>
    <label><input type="radio" name="gender" value="M"> Male</label>
    <label><input type="radio" name="gender" value="F"> Female</label><br>
    <button type="submit">Signup</button>
</form>
<?php if ($message !== ''): ?><p><?= e($message) ?></p><?php endif; ?>
</body></html>
```